

HCSM-43

Strong CP from the Orientation $\mathbb{Z}_2$

The Diagonal Flip as a Substrate Automorphism, the Exact Anticommutation $\{F, H_{\text{wind}}\} = 0$, and the Vanishing of θ_{eff}

Hodge Complex Standard Model

Stanley Preschutti

Entropia Research Institute / Information Physics Institute

ORCID: 0009-0004-5445-1744

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Abstract

The Hodge Complex Standard Model (HCSM) derives the Standard Model from the periodic 8×8 torus $\Lambda = \mathbb{Z}_8 \times \mathbb{Z}_8$. This paper closes the strong CP blocker of the framework. We construct the *orientation* $\mathbb{Z}_2$ of the substrate explicitly: it is the diagonal flip $F : (n_1, n_2) \mapsto (n_2, n_1)$, already defined in HCSM-00 Definition 5.3. On the 14-mode multiplet at $\lambda = 4$, F acts as an involution that preserves each D_4 orbit $\mathcal{O}_0, \mathcal{O}_1, \mathcal{O}_2$ individually and commutes with the discrete 5-point Laplacian. The central theorem of this paper is the *exact anticommutation*

$$\{F, H_{\text{wind}}\} = 0,$$

where $H_{\text{wind}} = i \sin(\pi M/8)$ is the winding generator of HCSM-00 Section 5.1. The anticommutation is not approximate: it follows analytically from the antisymmetry of $M_{ij} = n_1^{(i)} n_2^{(j)} - n_1^{(j)} n_2^{(i)}$ under the exchange of the two torus directions, and is verified at machine precision ($\max |\{F, H_{\text{wind}}\}| = 0.0$). Because F is an exact automorphism of the substrate Hamiltonian and the substrate vacuum lies in an F -eigensector, every F -odd observable has vanishing vacuum expectation value. The winding generator H_{wind} is F -odd; identifying it with the substrate realization of the topological density $F_{\mu\nu} \tilde{F}^{\mu\nu}$, the HCSM vacuum forces $\theta_{\text{QCD}} = 0$ exactly. The HCSM mass map produces positive diagonal masses, so $\arg \det(M_u M_d) = 0$ and therefore

$$\theta_{\text{eff}} = \theta_{\text{QCD}} + \arg \det(M_u M_d) = 0.$$

No axion, no Peccei-Quinn symmetry, no Nelson-Barr mechanism, and no new degrees of freedom are required. The strong CP problem is solved by the orientation $\mathbb{Z}_2$ of the substrate.

Derivation status summary. *Derived:* the orientation $\mathbb{Z}_2$ action F on the 14-mode multiplet; the orbitwise preservation of $\mathcal{O}_0, \mathcal{O}_1, \mathcal{O}_2$; the commutation $[F, \Lambda] = 0$; the exact anticommutation $\{F, H_{\text{wind}}\} = 0$; the vanishing vacuum expectation $\langle H_{\text{wind}} \rangle = 0$; the corollary $\theta_{\text{QCD}} = 0$; the corollary $\theta_{\text{eff}} = 0$. *Disclosed inputs:* physical identification $F \leftrightarrow \text{CP}$ (parallel to $A_1 \leftrightarrow \text{color}$, $B_1 \leftrightarrow \text{weak isospin}$); identification of H_{wind} with the substrate realization of $F_{\mu\nu} \tilde{F}^{\mu\nu}$; the HCSM mass map (HCSM-33). *Open:* whether the identification $F \leftrightarrow \text{CP}$ admits a substrate-level derivation (rather than a disclosed physical choice); the precise form of the EFT matching between H_{wind} and the standard $\theta F_{\mu\nu} \tilde{F}^{\mu\nu}$ operator.

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1 Introduction

1.1 The strong CP problem

Quantum chromodynamics admits a topological term

$$\mathcal{L}_\theta = \theta_{\text{QCD}} \frac{g_3^2}{32\pi^2} F_{\mu\nu}^a \tilde{F}^{a\mu\nu},$$

where $\tilde{F}^{a\mu\nu} = \frac{1}{2}\epsilon^{\mu\nu\rho\sigma}F_{\rho\sigma}^a$. The effective physical parameter is

$$\theta_{\text{eff}} = \theta_{\text{QCD}} + \arg \det(M_u M_d),$$

where M_u, M_d are the up- and down-quark mass matrices. The parameter θ_{eff} is CP-odd and is bounded by the neutron electric dipole moment [19],

$$|\theta_{\text{eff}}| \lesssim 10^{-10}.$$

The Standard Model offers no explanation for why θ_{eff} is so small: θ_{QCD} and $\arg \det(M_u M_d)$ are independent inputs, and their sum is not protected by any symmetry. This is the strong CP problem.

The standard solutions require either a new light pseudoscalar (the axion [14, 15, 16]), a parity-based mechanism with new heavy fermions (Nelson-Barr [17, 18]), or a spontaneously broken left-right symmetry. All of these extend the Standard Model with new degrees of freedom.

1.2 The HCSM setting

The Hodge Complex Standard Model derives the Standard Model from a single discrete substrate, the periodic 8×8 torus $\Lambda = \mathbb{Z}_8 \times \mathbb{Z}_8$, carrying the Hodge complex $\Omega^0 \oplus \Omega^1 \oplus \Omega^2$ and the Hodge–Dirac operator $D = d + d^*$ with $D^2 = \Delta$ exactly on each form degree [1, 3]. The 14 modes at the self-paired eigenvalue $\lambda = 4$ are identified with one Standard Model generation [4, 8]. The processor $T = I - P_C$ [5] acts on the 14-mode multiplet; its winding generator is

$$H_{\text{wind}} = i \sin\left(\frac{\pi}{8} M\right), \quad M_{ij} = n_1^{(i)} n_2^{(j)} - n_1^{(j)} n_2^{(i)}, \quad (1)$$

with spectrum $\{-4\sqrt{2}, -2\sqrt{2}, 0^{(10)}, +2\sqrt{2}, +4\sqrt{2}\}$ [6]. The matrix H_{wind} is Hermitian with entries in $\mathbb{Q}(\sqrt{2})$.

The framework already contains a natural candidate for a discrete orientation symmetry: the *diagonal flip*

$$F : (n_1, n_2) \mapsto (n_2, n_1),$$

introduced in HCSM-00 Definition 5.3 [1] and used in HCSM-40 [13] to characterize the self-mirror orbit $\mathcal{O}_2$. The zero state s is stated to be F -odd [7].

What has not been established is the *anticommutation* of F with H_{wind} . That is the missing ingredient. In this paper we prove it, and we show that it forces $\theta_{\text{eff}} = 0$ exactly.

1.3 Summary of results

1. (**Theorem 3.1**). F is an involution on the 14-mode multiplet, preserves each D_4 orbit $\mathcal{O}_0, \mathcal{O}_1, \mathcal{O}_2$ individually, and commutes with the discrete 5-point Laplacian Δ : $[F, \Delta] = 0$.
2. (**Theorem 4.1**). $\{F, H_{\text{wind}}\} = 0$ exactly, and $[F, H_{\text{wind}}] \neq 0$. The anticommutation follows analytically from the antisymmetry of M under the exchange of n_1 and n_2 .
3. (**Theorem 5.3**). The substrate vacuum lies in an F -eigensector, so every F -odd observable has vanishing expectation value: $\langle H_{\text{wind}} \rangle = 0$. Hence $\theta_{\text{QCD}} = 0$ exactly.
4. (**Corollary 6.2**). With the HCSM mass map [9], $\arg \det(M_u M_d) = 0$, so $\theta_{\text{eff}} = 0$ exactly.

1.4 Organization

Section 2 recalls the substrate, the 14-mode multiplet, and the winding generator. Section 3 constructs the orientation $\mathbb{Z}_2$ and proves the orbitwise preservation and commutation with the Laplacian. Section 4 proves the exact anticommutation. Section 5 derives $\theta_{\text{QCD}} = 0$. Section 6 derives $\theta_{\text{eff}} = 0$. Section 7 catalogues the derivation status. Section 8 states falsifiable predictions. Section 9 concludes.

2 Setup

2.1 The substrate and the 14-mode multiplet

The substrate is $\Lambda = \mathbb{Z}_8 \times \mathbb{Z}_8$ with $N = 64$ sites [2]. The discrete 5-point Laplacian is

$$(\Delta f)(x, y) = 4f(x, y) - f(x \pm 1, y) - f(x, y \pm 1),$$

with eigenvalues

$$\lambda(n_1, n_2) = 4 - 2 \cos\left(\frac{\pi n_1}{4}\right) - 2 \cos\left(\frac{\pi n_2}{4}\right), \quad n_1, n_2 \in \{0, \dots, 7\}.$$

The self-paired eigenvalue $\lambda = 4$ has multiplicity $2(L - 1) = 14$ [4], and the 14 modes fall into three D_4 orbits:

$$\mathcal{O}_0 = \{(0, 4), (4, 0)\}, \quad |\mathcal{O}_0| = 2, \quad (2)$$

$$\mathcal{O}_1 = \{(1, 3), (1, 5), (3, 1), (3, 7), (5, 1), (5, 7), (7, 3), (7, 5)\}, \quad |\mathcal{O}_1| = 8, \quad (3)$$

$$\mathcal{O}_2 = \{(2, 2), (2, 6), (6, 2), (6, 6)\}, \quad |\mathcal{O}_2| = 4. \quad (4)$$

2.2 The winding generator

The winding generator is defined by (1). Its matrix M is antisymmetric and real, so H_{wind} is Hermitian and traceless. The spectrum on the 14-mode subspace is

$$\sigma(H_{\text{wind}}) = \{-4\sqrt{2}, -2\sqrt{2}, 0^{(10)}, +2\sqrt{2}, +4\sqrt{2}\},$$

corresponding to winding classes A, B, C [6]. The class C of winding-zero modes has dimension 10 and coincides with $\ker H_{\text{wind}}$.

2.3 The diagonal flip

Definition 2.1 (Diagonal flip, HCSM-00 Def. 5.3). On mode labels, the diagonal flip is

$$F : (n_1, n_2) \mapsto (n_2, n_1).$$

The map F is an involution on $\mathbb{Z}_8 \times \mathbb{Z}_8$ and, as we shall see, restricts to the 14-mode multiplet.

2.4 Disclosed inputs

The framework's disclosed-input set [1] consists of three discrete items (Standard Model per-generation content; physical identifications $A_1 \leftrightarrow \text{color}$ and $B_1 \leftrightarrow \text{weak isospin}$; identification thesis 14 modes $\leftrightarrow$ 14 Standard Model particles) and one dimensionful anchor (M_P). This paper adds one further physical identification, stated explicitly in Section 5: the identification of F with the substrate realization of CP, and of H_{wind} with the substrate realization of $F_{\mu\nu} \tilde{F}^{\mu\nu}$. This is parallel in kind to the existing identifications and is a disclosed input, not a derivation.

3 The orientation $\mathbb{Z}_2$

3.1 Action on the 14-mode multiplet

Theorem 3.1 (Orientation $\mathbb{Z}_2$). *The diagonal flip F acts on the 14-mode multiplet at $\lambda = 4$ and satisfies:*

(i) $F^2 = I$ on the multiplet;

(ii) F preserves each D_4 orbit $\mathcal{O}_0, \mathcal{O}_1, \mathcal{O}_2$ individually;

(iii) $[F, \Delta] = 0$.

Proof. (i) and (ii) are direct. For (i), $F(F(m)) = m$ for every mode label m , hence $F^2 = I$.

For (ii), one checks each orbit:

$$\begin{aligned} F(\mathcal{O}_0) &= \{(4, 0), (0, 4)\} = \mathcal{O}_0, \\ F(\mathcal{O}_1) &= \{(3, 1), (5, 1), (1, 3), (7, 3), (1, 5), (7, 5), (3, 7), (5, 7)\} = \mathcal{O}_1, \\ F(\mathcal{O}_2) &= \{(2, 2), (6, 2), (2, 6), (6, 6)\} = \mathcal{O}_2. \end{aligned}$$

So each orbit is preserved setwise.

For (iii), write the Laplacian eigenvalue as $\lambda(n_1, n_2) = 4 - 2\cos(\pi n_1/4) - 2\cos(\pi n_2/4)$. This is manifestly symmetric in $n_1 \leftrightarrow n_2$, so $\lambda(F(m)) = \lambda(m)$ for every mode. Since Δ acts diagonally on the plane-wave basis with eigenvalue $\lambda(m)$, we have $[\Delta, F] = 0$ on the multiplet. $\square$

Remark 3.2. The orbitwise preservation of Theorem 3.1(ii) is why F is a *substrate* $\mathbb{Z}_2$ rather than a large reorganisation: it acts within each D_4 orbit, not across orbits. This is what permits it to be simultaneously an orientation flip and a symmetry of the D_4 orbit structure.

3.2 Fixed points

The fixed-point set of F on the 14-mode multiplet is

$$\text{Fix}(F) = \{(2, 2), (6, 6)\}.$$

These are exactly the two $\mathbb{Z}_2 \times \mathbb{Z}_2$ fixed points of [13, Theorem 7.1], identified there with the W boson $(2, 2)$ and the top quark $(6, 6)$. The fixed-point structure is thus *carried over* from the mirror orbit $\mathcal{O}_2 = \text{Fix}(M)$ established in HCSM-40; the diagonal flip does not introduce new fixed points.

3.3 The orbitwise sign structure

The numerical signed-permutation search on the 14-mode multiplet (Section A of the verification script) confirms two other commuting signed D_4 operators with nontrivial sign patterns on $(\mathcal{O}_0, \mathcal{O}_1, \mathcal{O}_2)$, in addition to F :

- the identity permutation with signs $(-1, -1, -1)$ (equivalent to $-I$ on the multiplet, trivially commuting with H_{wind});
- the 180 rotation $R_{180} : (n_1, n_2) \mapsto (-n_1, -n_2)$ with signs $(+1, -1, -1)$.

Both commute with H_{wind} ; the second will be exploited in HCSM-58 (joint independence of the geometric kernel). The present paper uses only F , which is the unique label-preserving $\mathbb{Z}_2$ that *anticommutes* with H_{wind} .

4 The exact anticommutation $\{F, H_{\text{wind}}\} = 0$

Theorem 4.1 (Anticommutation). *On the 14-mode multiplet at $\lambda = 4$,*

$$\{F, H_{\text{wind}}\} = 0 \quad \text{exactly,}$$

and

$$[F, H_{\text{wind}}] \neq 0.$$

Proof. Let $m_i = (n_1^{(i)}, n_2^{(i)}) \in \mathbb{Z}_8 \times \mathbb{Z}_8$ for $i = 1, \dots, 14$ be the 14 modes, and let p_F be the permutation of indices induced by F , i.e. $m_{p_F(i)} = F(m_i) = (n_2^{(i)}, n_1^{(i)})$.

The winding matrix is

$$M_{ij} = n_1^{(i)} n_2^{(j)} - n_1^{(j)} n_2^{(i)}.$$

Under F , both momenta are exchanged, so

$$M_{p_F(i) p_F(j)} = n_2^{(i)} n_1^{(j)} - n_2^{(j)} n_1^{(i)} = -(n_1^{(i)} n_2^{(j)} - n_1^{(j)} n_2^{(i)}) = -M_{ij}.$$

Thus, as a matrix identity on the multiplet,

$$P_F M P_F^{-1} = -M,$$

where P_F denotes the permutation matrix implementing F . Applying the odd function $x \mapsto i \sin(\pi x/8)$ elementwise,

$$P_F H_{\text{wind}} P_F^{-1} = P_F i \sin(\pi M/8) P_F^{-1} = i \sin(-\pi M/8) = -i \sin(\pi M/8) = -H_{\text{wind}}.$$

Equivalently, $P_F H_{\text{wind}} + H_{\text{wind}} P_F = 0$, which is $\{F, H_{\text{wind}}\} = 0$.

The commutator $[F, H_{\text{wind}}] = P_F H_{\text{wind}} - H_{\text{wind}} P_F = 2P_F H_{\text{wind}} \neq 0$, since $H_{\text{wind}} \neq 0$ and P_F is invertible. $\square$

Remark 4.2 (Analytic, not numerical). The proof uses only the antisymmetry of M under exchange of the two torus directions. It does not rely on any numerical diagonalization. The companion verification script confirms $\max |\{F, H_{\text{wind}}\}| = 0.0$ and $\max |[F, H_{\text{wind}}]| = 2.0$ in IEEE-754 double precision, exactly consistent with the analytic proof.

Remark 4.3 (The winding spectrum is F -paired). Since H_{wind} anticommutes with the invertible involution F , its spectrum is symmetric about zero: every nonzero eigenvalue λ pairs with $-\lambda$ via the F -conjugate eigenvector. This is a structural necessity given the spectrum $\{-4\sqrt{2}, -2\sqrt{2}, 0^{(10)}, +2\sqrt{2}, +4\sqrt{2}\}$ of HCSM-14. The 10-dimensional kernel is F -invariant.

5 Strong CP: $\theta_{\text{QCD}} = 0$

5.1 Physical identification

We identify the substrate operator H_{wind} with the realization of the topological density $F_{\mu\nu} \tilde{F}^{\mu\nu}$ in the HCSM effective description, and the diagonal flip F with the substrate realization of CP. This identification is a disclosed physical choice of the same character as $A_1 \leftrightarrow \text{color}$ and $B_1 \leftrightarrow \text{weak isospin}$: the substrate realizes a discrete symmetry, and its interpretation in Standard Model language is a physical identification, not a derivation.

The justification for this identification is threefold.

1. **Parity and CP action.** On the torus, the two fundamental cycles are exchanged by F . This is precisely what CP does to the two chiralities of a fermion. The discrete statement “exchange the two cycles” is the substrate counterpart of “exchange left and right.”
2. **The winding generator is the topological density.** On a two-torus, the topological density $F_{\mu\nu} \tilde{F}^{\mu\nu}$ reduces to the winding number density. The processor’s winding generator H_{wind} is the discrete realization of that density on the 14-mode multiplet.
3. **Analogy with HCSM-40.** In HCSM-40 [13], the same map F is used to characterize the mirror-fixed orbit $\mathcal{O}_2$. Its role there is structural (identifying fixed points); its role here is dynamical (forcing an operator to vanish on the vacuum). Both uses are consistent with the same disclosure.

5.2 The vacuum is in an F -eigensector

Lemma 5.1. *The substrate vacuum $|0\rangle$ lies in an eigensector of F with eigenvalue $\eta \in \{+1, -1\}$.*

Proof. By Theorem 3.1, F is an exact automorphism of the substrate Hamiltonian; in particular $[F, \Delta] = 0$. The vacuum is the ground state of the substrate's entropy functional, which is a functional of Δ alone. Any state that is an eigenstate of Δ can be chosen as an eigenstate of F , because F and Δ commute. The vacuum is therefore either F -even ($\eta = +1$) or F -odd ($\eta = -1$). $\square$

Remark 5.2. HCSM-22 [7] already states that the zero state s is F -odd; this is the substrate-level statement that the vacuum lies in the $\eta = -1$ eigensector. The choice of sign does not affect the corollaries below, which depend only on F being an eigensymmetry of the vacuum.

5.3 $\theta_{\text{QCD}} = 0$

Theorem 5.3 (Strong CP). *Every F -odd observable has vanishing vacuum expectation value. In particular,*

$$\langle 0 | H_{\text{wind}} | 0 \rangle = 0 \quad \text{exactly,}$$

and therefore

$$\theta_{\text{QCD}} = 0 \quad \text{exactly.}$$

Proof. Let $|0\rangle$ satisfy $F|0\rangle = \eta|0\rangle$ with $\eta = \pm 1$ (Lemma 5.1). Let $\mathcal{O}$ be any operator with $F\mathcal{O}F^{-1} = -\mathcal{O}$ (an F -odd operator). Then

$$\langle 0 | \mathcal{O} | 0 \rangle = \langle 0 | F^{-1}(F\mathcal{O}F^{-1})F | 0 \rangle = \eta^2 \langle 0 | (-\mathcal{O}) | 0 \rangle = -\langle 0 | \mathcal{O} | 0 \rangle,$$

so $\langle 0 | \mathcal{O} | 0 \rangle = 0$.

By Theorem 4.1, H_{wind} is F -odd. Therefore $\langle 0 | H_{\text{wind}} | 0 \rangle = 0$.

Under the physical identification of Section 5, H_{wind} is the substrate realization of $F_{\mu\nu}\tilde{F}^{\mu\nu}$ and the corresponding effective coupling is precisely θ_{QCD} . The vanishing of $\langle H_{\text{wind}} \rangle$ is therefore the statement that $\theta_{\text{QCD}} = 0$ exactly. $\square$

Remark 5.4 (No axion; no Peccei–Quinn). The mechanism is not spontaneous breaking of a Peccei–Quinn symmetry, and does not introduce a light pseudoscalar. It is a discrete symmetry of the substrate that fixes θ_{QCD} at zero. The standard axion mass formula $m_a \propto \sqrt{\chi}$ does not apply; there is no axion in the HCSM spectrum.

Remark 5.5 (No Nelson–Barr). The mechanism does not introduce new heavy fermions, and does not invoke a spontaneous CP-violating phase that is later cancelled. It uses only the discrete substrate symmetry F .

6 The effective θ_{eff}

6.1 The mass-matrix argument

The effective parameter is

$$\theta_{\text{eff}} = \theta_{\text{QCD}} + \arg \det(M_u M_d).$$

By Theorem 5.3, $\theta_{\text{QCD}} = 0$. It remains to show that $\arg \det(M_u M_d) = 0$ in the HCSM framework.

Lemma 6.1. *The HCSM quark mass matrices are positive diagonal in the mass-eigenstate basis, so $\arg \det(M_u M_d) = 0$ exactly.*

Proof. The HCSM mass map [9] assigns masses via

$$m_i = A_{\text{mass}} \exp(-B_{\text{mass}}(N/4)_i),$$

with $A_{\text{mass}} > 0$ and $B_{\text{mass}} > 0$ real. Every mass eigenvalue is therefore strictly positive. The mass matrices M_u , M_d , after diagonalization by the standard CKM procedure, are real positive diagonal. Hence $\det M_u > 0$ and $\det M_d > 0$, so $\arg \det(M_u M_d) = 0$. $\square$

6.2 Corollary: $\theta_{\text{eff}} = 0$

Corollary 6.2 (Vanishing θ_{eff}).

$$\theta_{\text{eff}} = \theta_{\text{QCD}} + \arg \det(M_u M_d) = 0.$$

Proof. Immediate from Theorem 5.3 and Lemma 6.1. $\square$

Remark 6.3. The argument is thus genuinely structural. It does not require fine-tuning between θ_{QCD} and $\arg \det(M_u M_d)$; both terms vanish separately. The neutron EDM bound $|\theta_{\text{eff}}| \lesssim 10^{-10}$ is satisfied trivially by $\theta_{\text{eff}} = 0$ exactly.

6.3 Consistency with the observed CP violation

The HCSM framework produces CP violation in the quark and lepton sectors via the CKM phase [10] and the Dirac PMNS phase [11]. The result $\theta_{\text{eff}} = 0$ is consistent with these: the CKM phase is a *flavour* phase, not a vacuum phase. The substrate $\mathbb{Z}_2$ forces the vacuum θ -angle to vanish while leaving the flavour structure untouched.

7 Derivation status

7.1 Derived from the substrate

1. The orientation $\mathbb{Z}_2$ action F on the 14-mode multiplet (Theorem 3.1).
2. The orbitwise preservation $F(\mathcal{O}_0) = \mathcal{O}_0$, $F(\mathcal{O}_1) = \mathcal{O}_1$, $F(\mathcal{O}_2) = \mathcal{O}_2$ (Theorem 3.1).
3. The commutation $[F, \Delta] = 0$ (Theorem 3.1).
4. The exact anticommutation $\{F, H_{\text{wind}}\} = 0$ (Theorem 4.1).
5. The vanishing of every F -odd vacuum expectation value (Theorem 5.3).
6. The corollary $\theta_{\text{QCD}} = 0$ (Theorem 5.3).
7. The corollary $\arg \det(M_u M_d) = 0$ (Lemma 6.1).
8. The corollary $\theta_{\text{eff}} = 0$ (Corollary 6.2).

7.2 Disclosed inputs

1. The physical identification $F \leftrightarrow \text{CP}$.
2. The physical identification of H_{wind} with the substrate realization of $F_{\mu\nu} \tilde{F}^{\mu\nu}$.
3. The HCSM mass map (HCSM-33).

These are disclosed in the same sense as the existing inputs of [1]: the substrate realizes a discrete structure, and its interpretation in Standard Model language is a physical identification. This paper does not claim to derive the identification from the substrate.

7.3 Open

1. **Substrate derivation of $F \leftrightarrow \text{CP}$.** The physical identification is currently a disclosed input. It would be stronger if the CP action could be derived from the substrate's own structure, e.g. from the intersection form on $\ker D$ or from the D_4 Schur orthogonality.
2. **EFT matching between H_{wind} and $\theta F_{\mu\nu} \tilde{F}^{\mu\nu}$.** The present derivation identifies H_{wind} with the topological density at the substrate level. A precise matching to the standard θ -term in the low-energy effective theory (with the correct normalization $g_3^2/32\pi^2$) remains to be written down. This is a normalisation problem, not a structural one; it does not affect the vanishing of θ_{QCD} , which follows from the F -oddness of H_{wind} .

8 Falsifiable predictions

1. $\theta_{\text{QCD}} = 0$ **exactly.** Any future measurement of a nonzero θ_{QCD} at any scale falsifies the framework.
2. **No axion.** The HCSM spectrum contains no light pseudoscalar. A confirmed axion detection at any mass would falsify the framework.
3. **No Nelson–Barr sector.** No new heavy fermions beyond the HCSM hidden content $5 + 6 + 31$ [12] are required or predicted.
4. **The neutron EDM vanishes identically.** The HCSM prediction is $d_n^{\text{QCD}} = 0$; any observed nonzero contribution to the neutron EDM from the QCD θ -term falsifies Theorem 5.3.
5. **F -oddness of the winding spectrum.** Any computational test that finds $\max |\{F, H_{\text{wind}}\}| > 0$ at machine precision falsifies Theorem 4.1. The companion verification script checks this at every commit.

9 Conclusion

We have closed the strong-CP blocker of the Hodge Complex Standard Model. The orientation $\mathbb{Z}_2$ of the substrate is the diagonal flip $F : (n_1, n_2) \mapsto (n_2, n_1)$ already defined in HCSM-00 Definition 5.3. On the 14-mode multiplet at $\lambda = 4$, F preserves each D_4 orbit individually and commutes with the discrete Laplacian. Its central structural property is the exact anticommutation

$$\{F, H_{\text{wind}}\} = 0,$$

with the winding generator $H_{\text{wind}} = i \sin(\pi M/8)$. This forces every F -odd observable to have vanishing vacuum expectation value. Since H_{wind} is F -odd and is identified with the substrate realization of $F_{\mu\nu} \tilde{F}^{\mu\nu}$, the HCSM vacuum forces $\theta_{\text{QCD}} = 0$ exactly. The HCSM mass map produces positive masses, so $\arg \det(M_u M_d) = 0$ as well. Therefore

$$\theta_{\text{eff}} = \theta_{\text{QCD}} + \arg \det(M_u M_d) = 0 \quad \text{exactly.}$$

No axion, no Peccei–Quinn symmetry, no Nelson–Barr sector, and no new degrees of freedom are introduced. The strong CP problem is solved by the substrate $\mathbb{Z}_2$.

The framework's strength is that it has fewer empirical anchors than the Standard Model, not that it has none.

Physics is what the machine cannot cancel. The Hodge complex is the machine.

A Reproducible verification

The companion script `HCSM-43_verification.py` verifies:

1. Enumeration of the 14 modes at $\lambda = 4$.
2. The orbit decomposition $\mathcal{O}_0, \mathcal{O}_1, \mathcal{O}_2$.
3. $F^2 = I$ on the multiplet.
4. $F(\mathcal{O}_0) = \mathcal{O}_0, F(\mathcal{O}_1) = \mathcal{O}_1, F(\mathcal{O}_2) = \mathcal{O}_2$.
5. $[F, \Delta] = 0$ at machine precision.
6. Construction of $H_{\text{wind}} = i \sin(\pi M/8)$ and verification of the spectrum $\{-4\sqrt{2}, -2\sqrt{2}, 0^{(10)}, +2\sqrt{2}, +4\sqrt{2}\}$.
7. $\max |\{F, H_{\text{wind}}\}| = 0.0$ at machine precision.
8. $\max |[F, H_{\text{wind}}]| = 2.0$ at machine precision.
9. $\text{Fix}(F) = \{(2, 2), (6, 6)\}$.
10. $\dim \ker H_{\text{wind}} = 10$.
11. $\langle s | H_{\text{wind}} | s \rangle = 0$ at machine precision.

Running `python HCSM-43_verification.py` reproduces every numerical claim.

B Numerical values

Quantity	Value
$\max \{F, H_{\text{wind}}\} $	0.0 (machine precision)
$\max [F, H_{\text{wind}}] $	2.0 (exact)
$\dim \ker H_{\text{wind}}$	10
$\text{Fix}(F)$	$\{(2, 2), (6, 6)\}$
$ \mathcal{O}_0 , \mathcal{O}_1 , \mathcal{O}_2 $	2, 8, 4
θ_{QCD} (HCSM)	0
$\arg \det(M_u M_d)$ (HCSM)	0
θ_{eff} (HCSM)	0
θ_{eff} observed bound	$\lesssim 10^{-10}$

C Web Crawler Metadata

C.1 Dublin Core

DC.title: HCSM-43: Strong CP from the Orientation $\mathbb{Z}_2$
DC.creator: Preschutti, Stanley
DC.creator.orcid: 0009-0004-5445-1744
DC.date.issued: 2026-09
DC.type: Text / Preprint / White Paper
DC.format: application/pdf; text/x-latex
DC.language: en
DC.publisher: Entropia Research Institute
DC.rights: All rights reserved
DC.subject: Hodge Complex Standard Model; HCSM; strong CP problem;

orientation $\mathbb{Z}_2$; diagonal flip; winding generator; anticommutation;
 theta_QCD; theta_eff; axion; neutron EDM
 DC.description: Explicit construction of the HCSM orientation $\mathbb{Z}_2$ as the
 diagonal flip F on the 14-mode multiplet. Theorem: $\{F, H_{\text{wind}}\}=0$ exactly.
 Corollary: $\theta_{\text{QCD}} = 0$ and $\theta_{\text{eff}} = 0$. No axion required.
 DC.identifier: HCSM-43-2026-09
 DC.source: <https://github.com/007STAN/HODGE-COMPLEX-STANDARD-MODEL-HCSM->

C.2 BibTeX

```
@article{PreschuttiHCSM43,
  author = {Preschutti, Stanley},
  title = {{HCSM-43}: Strong CP from the Orientation  $\mathbb{Z}_2$ },
  journal = {HCSM White Paper Series},
  year = {2026},
  month = {September},
  note = {ORCID: 0009-0004-5445-1744},
  keywords = {Hodge Complex Standard Model, HCSM, strong CP problem,
    orientation  $\mathbb{Z}_2$ , diagonal flip, winding generator, anticommutation},
  abstract = {Explicit construction of the HCSM orientation  $\mathbb{Z}_2$  as the
    diagonal flip  $F$  on the 14-mode multiplet. Theorem:  $\{F, H_{\text{wind}}\}=0$ 
    exactly. Corollary:  $\theta_{\text{QCD}} = 0$  and  $\theta_{\text{eff}} = 0$ . No axion.},
  url = {https://github.com/007STAN/HODGE-COMPLEX-STANDARD-MODEL-HCSM-}
}
```

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